

written by Joseph Popp. It was able to hide the file names on the hard drive and displayed a message claiming that the licence to use a certain piece of software had expired. The user was asked to pay a ransom amount of US\$ 189 to be paid to the PC Cyborg Corporation, in order to obtain a decrypting tool. Thanatos was another ransomware detected in the year 2018 and was the first to accept ransom payment in Bitcoins. This ransomware was able to encrypt the stored files on the hard drive; it changed the file extension to '.THANATOS' and demanded a ransom amount of 0.1 Bitcoin in order to decrypt the file.

- **Spyware:** This type of malware acts as a spy and gathers information about the user and the data stored in the computer. The gathered information is sent to the spyware developer and the infected computer is kept under surveillance. Such spyware can come from an unreliable third party software or even through a legitimate app. Some trusted websites can also run spyware when users visit them to keep track of them. The purpose of spyware can vary from advertising to hacking, and that's why it's difficult to determine whether spyware can be harmful or not.

Reader Rabbit, Internet Optimizer, CoolWebSearch, SpyEye and Go Keyboard are a few examples of spyware. Among these, Reader Rabbit was the first spyware detected in the year 2000 by ZoneAlarm – a firewall/antivirus software developed and maintained by Zone Labs. Reader Rabbit was educational software developed and marketed by the Mattel toy company. But this software was working as spyware and transmitted the user's information back to Mattel. The Go Keyboard app available in the Google Play store also works as spyware and transmits personal information to remote servers without the explicit consent of the user. This information includes the user's

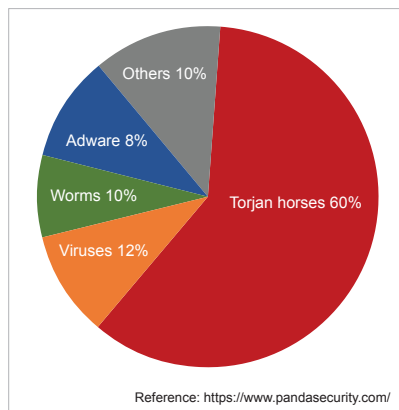


Figure 1: Malware category

Google account email, language, IMSI, location, network type, Android version and build, and the device's model and screen size. However, the developers of this app – a Chinese company called GOMO Dev Team – claim in the app's description that they will never collect personal data including credit card information.

- **Adware:** This type of malware doesn't do any harm to the data, but it displays advertisements to users through the user interface (UI). The purpose of this malware is to generate revenue for the developer in two ways – through the display of the advertisement and through 'pay-per-click', where revenue is generated based on the number of clicks by users on the displayed advertisement. Adware often comes with freeware (free to use apps) and this is the most common reason for a developer to offer an app for free. In the Google Play store, most of the free apps and games come with adware, and that's how both Google and the developer generate revenue. The most common problems from adware are frequent system crashes and slow Internet speed.
- **Rogue security software:** This is a kind of malware that displays a false virus alert to the user and asks for the purchase of a virus removal tool. This type of malware falls between scareware (malware that scares the user) and ransomware. In 2005,

SpySheriff rogue security software misled users with a false security alert and scared them to buy the software.

- **Wiper:** This type of malware is intended to wipe the hard drive of the user's computer. The attack on Sony Pictures in 2014 was done by a hacker group, which used a variant of the Shamoon wiper malware to erase the company's computer infrastructure.
- **Scareware:** This unique malware is intended to only scare the user by displaying false security alerts. However, some scareware can work as ransomware and demand the purchase of a tool that can resolve the security issue. Most scareware reaches users through social engineering and many untrusted websites can run scareware script also. In 2005, Microsoft and Washington State successfully sued Secure Computer (makers of Spyware Cleaner) for US\$ 1 million over charges of using scareware pop-ups. According to one report from Panda Security (a Spanish cybersecurity software company), the prevalence of malware by category is 60 per cent Trojans, 12 per cent viruses, 10 per cent worms, 8 per cent adware, and 10 per cent others.

The techniques used by antivirus or anti-malware software

Malware may be of various types and could give many kinds of problems. To get rid of any such malware, we need powerful antivirus software. An antivirus or anti-malware is software that is capable of preventing and detecting a malware, but may not always be able to remove it. It was first developed to detect only computer viruses because these were the more common types of malware attacks initially. That's how it got the name 'antivirus'. But nowadays, such antivirus software is also capable of dealing with many kinds of malware; so now such software can also be termed as anti-malware. Reaper was the first